

Exploratory Data Analysis (EDA) Report on Titanic Dataset

Task 5: Exploratory Data Analysis

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1. Introduction

Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset before advanced analysis. In this project, the Titanic dataset was analyzed using Pandas, Matplotlib, and Seaborn to discover patterns, trends, and relationships.

2. Objective

- Understand the dataset.
- Identify missing values and duplicates.
- Clean the data.
- Perform visualizations.
- Discover patterns and insights.

3. Dataset Information

Dataset: Titanic Dataset (Kaggle)

Records: 891

Features: 12

4. Libraries Used

Pandas, Matplotlib, Seaborn, Jupyter Notebook (VS Code)

5. Data Understanding

Explored the dataset using `head()`, `info()`, `describe()`, `shape`, `columns`, `duplicated()`, and `isnull()`. Missing values were found in Age (177), Cabin (687), and Embarked (2). No duplicate records were found.

6. Data Cleaning

Removed the Cabin column because about 77% of its values were missing. Filled missing Age values using the median and missing Embarked values using the mode. Verified that no missing values remained.

7. Exploratory Data Analysis

Performed Univariate Analysis (Survival, Gender, Passenger Class, Age Distribution, Box Plot), Bivariate Analysis (Survival vs Gender, Passenger Class, Age, Fare), and Multivariate Analysis (Correlation Heatmap and Pair Plot).

8. Key Findings

- Most passengers did not survive.
- Most passengers were male.
- Most passengers traveled in Third Class.
- Female passengers had a higher survival rate than male passengers.
- First Class passengers had a higher survival rate than Third Class passengers.
- Higher ticket fares were generally associated with higher survival.
- Passenger Class and Fare showed a noticeable relationship.

9. Conclusion

The analysis revealed that gender, passenger class, and fare significantly influenced survival. Data cleaning improved the quality of the dataset, and visualizations transformed raw data into meaningful insights.

10. Tools Used

Python, Pandas, Matplotlib, Seaborn, VS Code Jupyter Notebook

11. Outcome

This project demonstrates a complete Exploratory Data Analysis workflow from data understanding and cleaning to visualization and business insight generation.